

Adenocarcinoma of the Esophagus and GE Junction

1

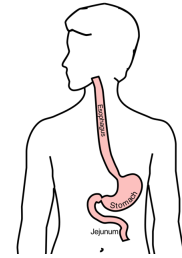
Anatomy

Food moves from the throat

→ esophagus

→ stomach

→ small bowel (jejunum)



2

Cancer Staging

Staging refers to the tests to determine

- How large is the tumor?
- Has there been spread to lymph nodes?
- Has it spread to other parts of the body?

Treatment options depend upon the cancer stage

3

Cancer Staging

• **T** = Tumor - Depth of growth into the wall

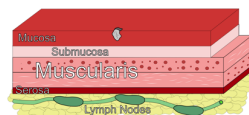
• **N** = Nodes - Spread to the lymph nodes

• **M** = Metastasis - Spread to liver, lungs, or bone

4

Early Stage Cancers

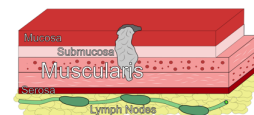
Cancers start on the very inside layer called the mucosa



5

Locally-advanced Cancers

Over time, cancers can grow into the muscular wall



6

Lymph Nodes

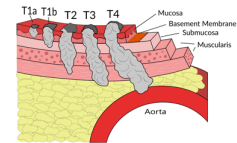
In some cases, cancer cells can break off from the main tumor and spread to lymph nodes



7

T Stage

Cancers are categorized based upon the thickness of the tumor, known as the T stage

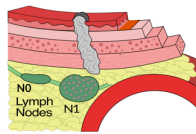


8

N Stage

Cancers are categorized by whether there is spread to the nodes.

- **N0** cancers have not spread to the nodes
- **N1** cancers have spread to the nodes.

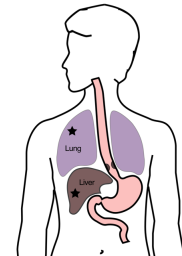


9

M Stage

Some cancers spread to other parts of the body

- **M0** cancers have not spread
- **M1** cancers have spread to lungs, liver, bone, or within the abdomen



10

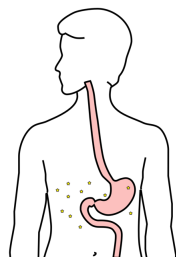
Carcinomatosis (M1)

Spread within the abdomen is

carcinomatosis

Tumors can be very small (grain of rice)

Fluid can also build up: **ascites**



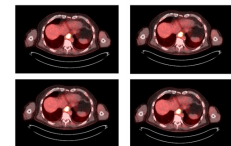
11

PET scan

Similar to CT scan

Tracer shows 'hot spots'

- Cancer
- Inflammation or infection
- Normal organs (heart, kidneys)



12

PET scan Preparation

- Nothing to eat or drink for 4 hours prior to exam
- High-protein low-carbohydrate diet the day prior
- No strenuous activity day prior to study

13

PET scan Preparation - Diabetic Patients

PET scan uses a modified sugar as a tracer $\Rightarrow$ you will need to be careful about blood sugar and insulin before the study

- No insulin within 4 hours of the exam
- OK to take diabetes pills the day of the exam
- PET scans are generally best in the morning
- Plan to eat a meal after the PET scan, with half the usual morning dose of insulin

14

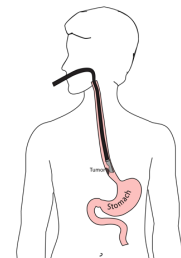
PET scan Preparation - Diabetic Patients

- If PET is scheduled after 10AM:
 - low-carbohydrate breakfast at least 4 hours before
 - half the usual regular (short-acting) insulin
- No long-acting insulin after midnight.
- Insulin pump: turn it off 4 hours before the study (or turn it down to basal/night-time rate)

15

Endoscopic Ultrasound

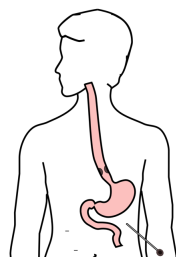
- Similar to upper endoscopy (EGD)
- Ultrasound in scope
- Evaluates T stage



16

Laparoscopy

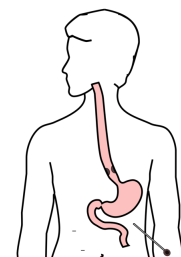
- Some stomach cancers can spread inside the abdomen
- Areas of spread can be very small (grain of rice)
- Laparoscopy can detect spread inside the abdomen



17

Laparoscopy

- General anesthetic
- 1/4" incisions
- Telescope examines the abdomen
- Biopsies performed
- Fluid sampled



18

Treatment Plan

Superficial (T1)

- Endoscopic Therapy

Localized (T1b/T2)

- Surgery (esophagectomy)

Locally-advanced (T3M0)

- Chemo±Radiation → Surgery

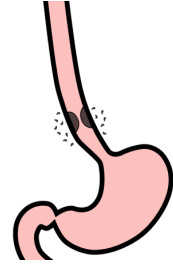
Metastatic (M1)

- Chemotherapy

19

Locally-advanced cancers

Patients with locally-advanced esophageal cancer often have localized spread of cancer cells in the surrounding area



20

Locally-advanced cancers

If locally-advanced cancers are treated with surgery alone...



21

Locally-advanced cancers

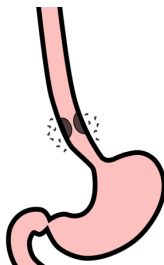
If locally-advanced cancers are treated with surgery alone...
There is a risk that cancer cells can be left behind



22

Preoperative Therapy

It is helpful to start with therapy *before* surgery that will shrink the cancer.



23

Preoperative Therapy

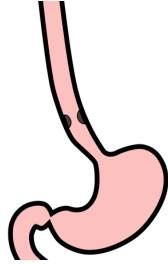
It is helpful to start with therapy *before* surgery that will shrink the cancer.



24

Preoperative Therapy

It is helpful to start with therapy *before* surgery that will shrink the cancer.



25

Surgery after Preoperative Therapy

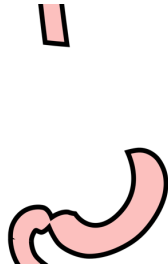
When surgery is then performed...



26

Preoperative Therapy

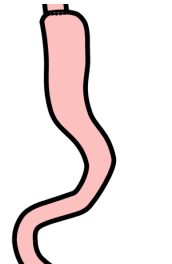
When surgery is then performed...
The risk of cancer recurrence is minimized



27

Preoperative Therapy

When surgery is then performed...
The risk of cancer recurrence is minimized



28

Chemotherapy + Radiation CROSS Trial

363 patients with esophageal cancer studied
Patients were treated in two groups:

Surgery Alone

vs

Chemo + Radiation → Surgery

29

Chemotherapy + Radiation CROSS Trial

363 patients with esophageal cancer studied
Chemotherapy + radiation together over 6 wks

Surgery Alone

vs

Chemo + Radiation → Surgery ⇒ Longer Survival

30

Chemo + Radiation CROSS Trial

Typical schedule for chemotherapy + radiation:

- Chemotherapy once per week for six weeks
- Radiation five days per week for six weeks (28)
- PET scan (or CT) 4 weeks after the end of radiation
- Surgery 8 weeks after the end of radiation

31

Chemo + Radiation - Side Effects

Kills cancer cells in the esophagus and lymph nodes

Can also irritate the lining of the esophagus.

Swallowing can be difficult the last 2 weeks.

Feeding tube may be needed for hydration/nutrition.

32

Locally-advanced Adenocarcinoma

“Sandwich” chemotherapy before + after surgery:

Chemo (8 wks) → Surgery → Chemo (8 wks)

Two different drug combinations:

- FLOT (more effective)
- FOLFOX (better tolerated)

33

“Sandwich” Chemotherapy Drugs

FLOT

- 5-FU
- Leucovorin
- Oxaliplatin
- Taxotere

FOLFOX

- 5-FU
- Leucovorin
- Oxaliplatin

34

FLOT Treatment Plan

- FLOT Chemo every 2 weeks x 4 (=8 weeks total)
- Surgery (4-6 weeks later)
- FLOT Chemo every 2 weeks x 4 (=8 weeks total)

35

Durvalumab Immunotherapy

Cancer cells can turn off immune cells using a protein PD-L1

Durvalumab turns on immune cells by counteracting PD-L1

Durvalumab strengthens the immune system to fight cancer

36

Matterhorn Trial

The Matterhorn clinical trial compared two types of treatment:

FLOT → Surgery → FLOT

FLOT + Durvalumab → Surgery → FLOT + Durvalumab

Better survival with addition of durvalumab to FLOT

Treatment with durvalumab depends upon approval from insurance company

37

FLOT Chemo ± Durvalumab

FLOT Chemotherapy

- FLOT (8 weeks)
- Surgery (4-6 weeks later)
- FLOT (8 weeks)

FLOT + Durvalumab

- FLOT + Durvalumab (8 wks)
- Surgery (4-6 weeks later)
- FLOT + Durvalumab (8 wks)
- Durvalumab monthly

38

Tumor Biomarkers - Pathology Tests

Show whether other drugs may be helpful:

- HER-2 → Herceptin can be helpful
- PD-L1 → Immunotherapy can be helpful
- MMR → Immunotherapy can be very helpful

Biomarkers reported in a separate pathology report

Your medical oncologist will review these with you

39

Adenocarcinoma Treatment Options

Chemo + Radiation

- Chemo + Radiation (6wk)
- Surgery

Chemotherapy

- Chemo (8wk)
- Surgery
- Chemo (8wk)

40

Adenocarcinoma Treatment Options

Chemo + Radiation

- Longer track record
- Better tolerated
- Port usually placed
- Eating worse → better
- May need feeding tube

Chemotherapy

- More effective
- More side effects
- Port always required
- Eating slowly improves
- Less likely to need feeding tube

41

Chemotherapy Administration

Most chemotherapy is administered by vein.

Several options exist to administer chemotherapy:

- Intravenous catheter in peripheral veins
- Peripheral Intravenous Central Catheter (PICC)
- Central Venous port

42

Intravenous Catheter in Peripheral Vein ("IV")

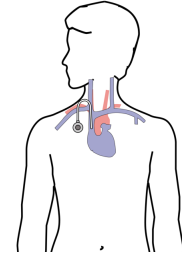
- IV catheter placed in vein of hand or arm
- Allows administration of chemo and fluids
- Placed for each dose
- Removed that day
- Not suitable for FLOT chemo



43

Central Venous Port

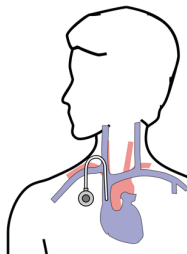
- Implantable device makes chemo easier
- May shower in 24 hrs
- No special care at home
- OK for FLOT chemo
- Allows for blood draws



44

Central Venous Port

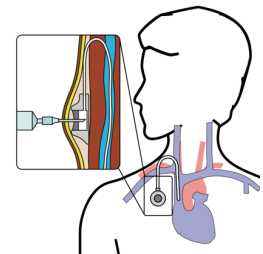
- Implanted under skin
- Neck incision (1/4")
- Incision below the collarbone
- Sutures dissolve
- "Superglue" on incisions



45

Central Venous Port

When it is time for chemotherapy, a needle is inserted through the skin into the port



46

Restaging

CT or PET scan will be performed after preoperative therapy

- Surgery performed after restaging
- Timing depends upon recovery from therapy

47

Primary Care Practitioner (PCP)

Critical to coordinate care between specialists.
We will update your PCP after each visit
PCP Referral Line (844) 235-6998

48

My Atrium Patient Portal

- Critical to good communication with your care team
- Available for desktop or laptop or phone
- Sign up at my.atriumhealth.org

49

Exercise

- Reduces risk of complications from treatment
- Goal is 30min/day of vigorous exercise 6 days/wk
 - Working hard enough that you can't converse
 - Start slowly and build up
 - Every day counts! (Aim for daily activity)

50

Smoking Cessation

- Smoking makes cancer treatment more difficult
 - Increases risk of complications after surgery
- Options for help with smoking cessation:
 - NC Quit Line 1-800-QUIT-NOW (1-800-784-8669)
 - American Lung Assn
www.freedomfromsmoking.org
 - Smoking Cessation Counseling (Metro Charlotte)

51

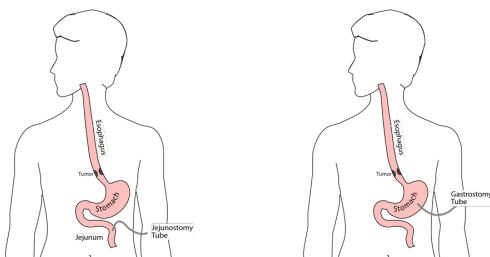
Protein Needs

- Men: Average 75 grams/day
 - Women: Average 60 grams/day
- Protein Shakes provide protein with minimal sugar



52

Feeding Tubes



53

Gastrostomy Tube

Feeding Gastrostomy

- Feeding with a syringe several times per day.
- Tube can be hidden underneath clothing
- Tube does not interfere with eating by mouth
- Removed easily in the office when no longer needed

54

Gastrostomy Tube Methods

PEG: Tube placed by endoscopy

Laparoscopic: Tube placed surgically by laparoscopy

Preferred method depends upon whether esophagectomy is planned

55

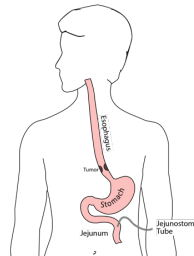
Gastrostomy Tube

- Outpatient Placement (go home the same day)
- Central venous port can be placed at the same time (if needed)

56

Jejunostomy Tube

- Nutrition to bypasses the esophagus and stomach
- Placed in small intestine
- Pump administers feedings slowly
- Feeding done at night



57

Jejunostomy Typical Regimen

- Jejunostomy tube feeds for 16 hours (6pm-10am)
 - Men: 75mL/hour x 16 hours = 5 cartons
 - Women: 60mL/hour x 16 hours = 4 cartons
 - Water 240ml (8oz) via syringe 4x/day
- Hospital nurses will teach use of the feeding tube

58

Jejunostomy Feeds with Diabetes

Jejunostomy feedings elevate blood sugars

- Insulin may be required along with feeds
- Typical Pattern for tube feeds
- Feeds run via pump from 6pm to 10am
 - Insulin at 6pm (70/30 insulin)
 - Insulin at Midnight (70/30 insulin)
 - No insulin if tube feedings are not run

59

Jejunostomy Video

A video is available to help become familiar with the feeding jejunostomy



60